

Research Notes on Receiver operating characteristic

Receiver operating characteristic

>> Section 1

Z-score If a standard score is applied to the ROC curve, the curve will be transformed into a straight line. This z-score is based on a normal distribution with a mean of zero and a standard deviation of one. In memory strength theory, one must assume that the zROC is not only linear, but has a slope of 1.0. The normal distributions of targets (studied objects that the subjects need to recall) and lures (non studied objects that the subjects attempt to recall) is the factor causing the zROC to be linear. The linearity of the zROC curve depends on the standard deviations of the target and lure strength distributions. If the standard deviations are equal, the slope will be 1.0. If the standard deviation of the target strength distribution is larger than the standard deviation of the lure strength distribution, then the slope will be smaller than 1.0. In most studies, it has been found that the zROC curve slopes constantly fall below 1, usually between 0.5 and 0.9. Many experiments yielded a zROC slope of 0.8. A slope of 0.8 implies that the variability of the target strength distribution is 25% larger than the variability of the lure strength distribution. Another variable used is d' (d prime) (discussed above in "Other measures"), which can easily be expressed in terms of z-values. Although d' is a commonly used parameter, it must be recognized that it is only relevant when strictly adhering to the very strong assumptions of strength theory made above. The z-score of an ROC curve is always linear, as assumed, except in special situations. The Yonelinas familiarity-recollection model is a two-dimensional account of recognition memory. Instead of the subject simply answering yes or no to a specific input, the subject gives the input a feeling of familiarity, which operates like the original ROC curve. What changes, though, is a parameter for Recollection (R). Recollection is assumed to be all-or-none, and it trumps familiarity. If there were no recollection component, zROC would have a predicted slope of 1. However, when adding the recollection component, the zROC curve will be concave up, with a decreased slope. This difference in shape and slope result from an added element of variability due to some items being recollected. Patients with anterograde amnesia are unable to recollect, so their Yonelinas zROC curve would have a slope close to 1.0.

>> Section 2

History The ROC curve was first used during World War II for the analysis of radar signals before it was employed in signal detection theory. Following the attack on Pearl Harbor in 1941, the United States military began new research to increase the prediction of correctly detected Japanese aircraft from their radar signals. For these purposes they measured the ability of a radar receiver

operator to make these important distinctions, which was called the Receiver Operating Characteristic. In the 1950s, ROC curves were employed in psychophysics to assess human (and occasionally non-human animal) detection of weak signals. In medicine, ROC analysis has been extensively used in the evaluation of diagnostic tests. ROC curves are also used extensively in epidemiology and medical research and are frequently mentioned in conjunction with evidence-based medicine. In radiology, ROC analysis is a common technique to evaluate new radiology techniques. In the social sciences, ROC analysis is often called the ROC Accuracy Ratio, a common technique for judging the accuracy of default probability models. ROC curves are widely used in laboratory medicine to assess the diagnostic accuracy of a test, to choose the optimal cut-off of a test and to compare diagnostic accuracy of several tests. ROC curves also proved useful for the evaluation of machine learning techniques. The first application of ROC in machine learning was by Spackman who demonstrated the value of ROC curves in comparing and evaluating different classification algorithms. ROC curves are also used in verification of forecasts in meteorology.

>> Section 3

ROC curves beyond binary classification The extension of ROC curves for classification problems with more than two classes is cumbersome. Two common approaches for when there are multiple classes are (1) average over all pairwise AUC values and (2) compute the volume under surface (VUS). To average over all pairwise classes, one computes the AUC for each pair of classes, using only the examples from those two classes as if there were no other classes, and then averages these AUC values over all possible pairs. When there are c classes there will be $c(c - 1) / 2$ possible pairs of classes. The volume under surface approach has one plot a hypersurface rather than a curve and then measure the hypervolume under that hypersurface. Every possible decision rule that one might use for a classifier for c classes can be described in terms of its true positive rates ($TPR_1, \dots, TPR_c$). It is this set of rates that defines a point, and the set of all possible decision rules yields a cloud of points that define the hypersurface. With this definition, the VUS is the probability that the classifier will be able to correctly label all c examples when it is given a set that has one randomly selected example from each class. The implementation of a classifier that knows that its input set consists of one example from each class might first compute a goodness-of-fit score for each of the c^2 possible pairings of an example to a class, and then employ the Hungarian algorithm to maximize the sum of the c selected scores over all $c!$ possible ways to assign exactly one example to

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each class. Given the success of ROC curves for the assessment of classification models, the extension of ROC curves for other supervised tasks has also been investigated. Notable proposals for regression problems are the so-called regression error characteristic (REC) Curves and the Regression ROC (RROC) curves. In the latter, RROC curves become extremely similar to ROC curves for classification, with the notions of asymmetry, dominance and convex hull. Also, the area under RROC curves is proportional to the error variance of the regression model.